

LOW-CODE CHATBOT BUILDER

✦ **B.TECH PROGRESS PRESENTATION: PHASE-2** ✦
(AIC4990)

TEAM

IMAAD HASAN

SAYYED ARIB HUSSAIN

SUPERVISED BY : DR. NADEEM AKHTAR



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TERMINOLOGIES

- **Large Language Models(LLMs):** Large Language Models are advanced AI systems based on transformer architecture, designed to understand and generate human language. Utilizing deep neural networks trained on extensive datasets.
- **Chatbot:** A chatbot is a computer program that simulates human conversation through text or voice interactions. Integrated into messaging applications and websites, it automates responses to user inquiries, enhancing customer service and engagement.
- **Vector Database:** A vector database is a specialized database that stores data as high-dimensional vectors, allowing for efficient indexing and searching of unstructured data like text and images. It facilitates similarity searches based on vector embeddings, which represent the features of data objects numerically.
- **Retrieval-Augmented Generation(RAG):** RAG is a method that combines retrieval-based and generative approaches in natural language processing. It retrieves relevant documents from a database to provide context for generating more accurate and informative responses in conversational AI applications.
- **Langchain:** LangChain is an open-source framework that enables developers to create applications using large language models (LLMs) in combination with external data sources. It simplifies the integration of LLMs for building advanced natural language processing applications.

PROBLEM STATEMENT

- **Complexity of Chatbot Development**

Designing a chatbot is time-consuming and requires expertise in areas like RAG, Embeddings, vector databases and LLM frameworks.

- **Challenges of Existing Platforms**

Current chatbot builders have overwhelming UIs, features and lack reliable function calling support, complicating the creation of chatbots.

- **Accessibility Issues**

The specialized knowledge required and limitations of existing platforms create barriers for small businesses and individual developers, restricting access to chatbot technology.

LITERATURE SURVEY

Research/Title	Year	Authors	Approach	Key Contributions	Peformance/ Outcomes
Automated The Development of Task- Oriented LLM-Based Chatbots[1]	2024 (ACM)	Jesús Sánchez Cuadrado, J uan de Lara et al.	YAML-based DSL for LLM-based task-oriented chatbots, compiled into structured LLM prompts using the ReAct framework.	• Simplified the specs. of a custom chatbot. • Modular Design	Chatbots using this approach had smaller specs, better conversational flexibility, and more diverse outputs than intent-based chatbots, leading to an improved user experience

Research/Title	Publishing Year(with Publisher)	Authors	Approach	Key Contributions	Peformance/ Outcomes
Efficiency-Driven Custom Chatbot Development: Unleashing LangChain, RAG, and Performance-Optimized LLM Fusion[4]	2024 (Science Direct)	S. Vidive lli et al.	Implemented a two-dimensional data retrieval system combining PEFT and quantization to enhance resource efficiency in chatbot development. By integrating LangChain and Retrieval Augmented Generation (RAG) , it improves response accuracy and access to relevant medical data .	- Employed PEFT and quantization, enabling effective operation in resource-constrained environments. - Evaluation of chatbot performance using BLEU, ROUGE scores, user satisfaction surveys, and expert feedback.	- Base Model: - BLEU Score: 60 - ROUGE Score: 55 - Fine-Tuned Model: - BLEU Score: 72 - ROUGE Score: 68 RAG Implemented Model: BLEU Score: 83 ROUGE Score: 91 - Latency Metrics in case of RAG: - TTFT: 1.5 seconds - TPOT: 0.2 seconds - Overall Latency: ~3 seconds for complete response.

No.	Research/Title	Publishing Year(with Publisher)	Authors	Approach	Key Contributions	Performance/ Outcomes
3.	Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question Answering[3]	2024 (ACM)	Zhentao Xu, Mark Jerome Cruz et al.	Combined RAG with a Knowledge Graph built from customer service data to enable context-aware responses by retrieving relevant sub-graphs.	- Introduces the integration of KG into the RAG model, improving retrieval accuracy and context retention. - Enhances answer quality by leveraging structured data from historical customer service issues. - Demonstrates practical deployment within LinkedIn, showcasing the effectiveness of the approach.	- Outperformed baseline models by 77.6% in MRR and a 0.32 increase in BLEU score. Real-world deployment resulted in a 28.6% reduction in median issue resolution time.
4.	LOCALRQA	2024 (ACL)	Xiao Yu, Yunan Lu et al.	Provides an open-source toolkit for RQA systems, enabling <q, a, p> data generation and local deployment via evaluation pages or chat interfaces.	High Performance: 7B-models match OpenAI’s text-ada-002 & GPT-4-turbo. User-Focused: Supports data generation, human feedback, and local deployment.	Competitive Performance: Locally trained 7B-models achieve results comparable to leading commercial models. - User Feedback Integration: Enables real-time improvements via interactive chat interfaces. Efficient Deployment: Local hosting reduces latency, enhancing user experience.

PLAN

PHASE-I

OVERALL IDENTIFICATION OF EXISTING CHALLENGES

- Generalization vs Specialization
- Explanability and Transparency
- Evaluation Metrics[2]

DATASET

BiText[3].

- Spans across 27 different domains of customer chatbot qa pairs.
- 27,000 QA pairs (around 3.57 million tokens)
- Multilingual Dataset
- Train Dataset - 18900 QA pairs
- Test Dataset - 5400 QA
- Validation Dataset* - 150 QA

DEVELOP AND EVALUATE RAG

- Develop a RAG architecture using Langchain, a Vector DB and an LLM.
- Eval Metrics to be used: Accuracy, ROGUE Scores and Semantic Score, G- Eval.

PLAN

PHASE-II

EXPAND DATA SOURCE/IMPROVE RAG

- Improve proposed RAG Architecture(exploring Agentic RAG).
- Combine evaluation metrics (e.g., ROUGE and semantic-based metrics[4]) for comprehensive analysis.
- Expand dataset acquisition: Extract data from more text, images, and videos, scrape URL content, and process information from images.

RAG EVALUATION PIPELINE

- Evaluate 12 more datasets with approximately 25,000 rows each to evaluate our RAG, which amounts to at least 75M tokens.
- To evaluate this huge dataset using locally installed LLMs.
- Scrap data from the link augment question/answer pairs and pass on to knowledge base

DESIGN/DEVELOP WEB-APP & BUILD-TEST-DEPLOY

- System Design - UML Diagram, Database Design
- Create UI and server using NextJS and **FastAPI** respectively.
- Allow Integration with third-party applications.
- Deploy on AWS, monitor/improve performance in production.
- Continous Improvement..

PROGRESS SO FAR...

RAG TECHNIQUES EXPLORED:

- **Fine-tuning using language models** (e.g., Google Flan T-5 Large) and large language models (e.g., Llama 3 with 3.2 billion parameters) utilizing PEFT and QLoRA techniques.
- **Explored various RAG approaches** (from Naive RAG to Advanced RAG architecture) inference.

TECH STACK:

- LLMs from OpenAI GPT (via OpenAI API), Llama, Mixtral.
- Hugging Face Transformers
- Langchain
- Vector DB : Pinecone
- Frontend & Backend - FastAPI
 - & NextJS
- Web Scraping - BeautifulSoup

HARDWARE REQUIREMENTS:

- **CPU: Intel i7 13th Gen**
- **GPU: NVIDIA RTX 3060Ti**
- **RAM: 48GB DDR4**

EVALUATION METRICS

Model Name	Accuracy (%)	ROUGE-1	ROUGE-2	SEM SCORE	G-EVAL	ROUGE-SEM
LoRA Fine-tuned Language Model(Google Flan-T5)	44	0.44	0.18	0.28	1.2	0.53
PEFT & QLoRA Fine-Tuned Llama 3 Model (3.2B)[On HF]	50	0.43	0.16	0.59	2.21	0.57
Naive RAG	54	0.41	0.21	0.7	2.6	0.62
Fine-tuned RAG	66	0.45	0.24	0.77	2.6	0.66
RAG Fusion	74	0.48	0.24	0.80	3.0	0.70
Fine Tuned RAG Fusion	93	0.58	0.29	0.91	4.0	0.69

- INSIGHTS
- RAG offers the best balance of performance and scalability.
 - Fine Tuned RAG Fusion prioritizes understanding user intent (higher SEM score) over exact phrase matching.[5]
 - Optimized for real-world QA, focusing on meaningful responses over n-gram overlaps.

UNDERSTANDING LOW ROUGE SCORES IN RAG MODELS

Exact Lexical Matching:

- Traditional ROUGE metrics, including ROUGE-1, ROUGE-2, and ROUGE-L, **focus on exact lexical matches between reference and generated texts**, which can be limiting.[4]

Impact on LLM-Generated Outputs:

- Advanced Language Models (LLMs) often generate responses that **include synonyms and rephrased ideas**, which leads to lower ROUGE scores despite high-quality content.
- **In the example in Fig.1[5]**, despite being correct in meaning, traditional ROUGE metrics assign a **low score**.
- In our evaluation metric, we measure **Accuracy (LLM as a judge using the latest GPT-4.1 Model)** instead of **Human Judgment**, which evaluates generated content more closely to human perception, capturing semantic similarity.[6]

Context: ... After Peter returns, they eventually figure out her proper care, right down to diaper changes, baths, and feedings. The next day, **two men (who are drug dealers)** arrive at the apartment to pick up the package...

Question: Who comes to pick up the package the next day?

Gold Answers: **drug dealers, the drug dealer**

Prediction: **two men**

Human Judgement: 5 out of 5

ROUGE-L: 0

METEOR: 0

(a) Example from the generative **NarrativeQA** dataset.

Fig.1

Need for Enhanced Evaluation Metrics:

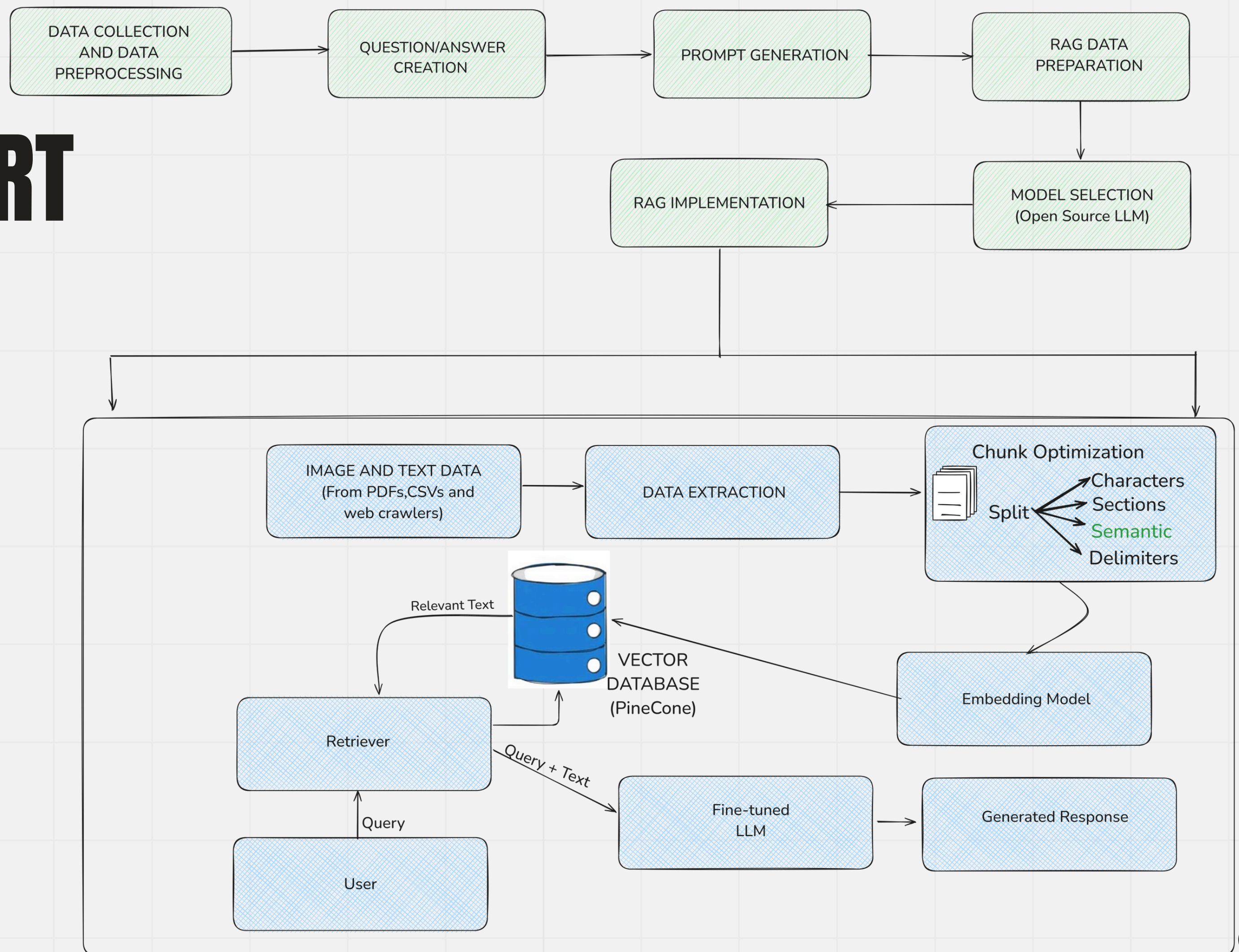
- **Lexical Overlap Limitations:** ROUGE, while useful, primarily focuses on word-level overlap, potentially overlooking semantically equivalent summaries with different phrasing.
- Therefore, moving beyond lexical-based metrics like ROUGE is essential. Metrics like **ROUGE-SEM** ensure a more comprehensive evaluation, accounting for **both lexical and semantic correctness**. [4]

Source document	Reference summary	Candidate summary	Lexically similar	Semantically similar	Categories	Back-translation	ROUGE-1/2/L	ROUGE-SEM-1/2/L
Dan: Buy me a sandwich on your way to work. Kevin: Ok, no problem! Dan: thanks!	Dan wants Kevin to buy him a sandwich on his way to work.	Kevin will buy Dan a sandwich on his way to work.	✓	✓	Good-summary		83.33/54.55/78.72	95.83/62.73/90.53
Frank: Hey! Hope: Hi! Frank: I love you. Hope: I love you too. Frank: Well, I hope so! Hope: Frankly, I really do.	Frank and Hope love each other.	Frank and Hope adore one another.	✗	✓	Pearl-summary	Frank and Hope adore each other. 83.33/60.00/85.90	50.00/40.00/56.12	93.75/45.00/63.14
Jesse: Can I borrow your razor? Stig: What happened to yours? Jesse: I broke it, fell right out of my hands.	Jesse broke his razor and wants to borrow Stig's.	Jesse broke Stig's razor and wants to borrow it.	✓	✗	Glass-summary	Stig's razor was broken and Jesse is requesting to use it. 60.00/0.00/36.69	88.89/62.50/81.10	33.50/0.00/20.48
Georgia: What do you think of this photo? Roxana: Buy it! Summer: You look great! Georgia: I like it, but where will i wear it? Summer: Parties.	Georgia sent a photo. Roxana and Summer advise Georgia to buy it.	Georgia is going to wear it to parties.	✗	✗	Bad-summary		30.00/0.00/36.77	28.5/0.00/34.93

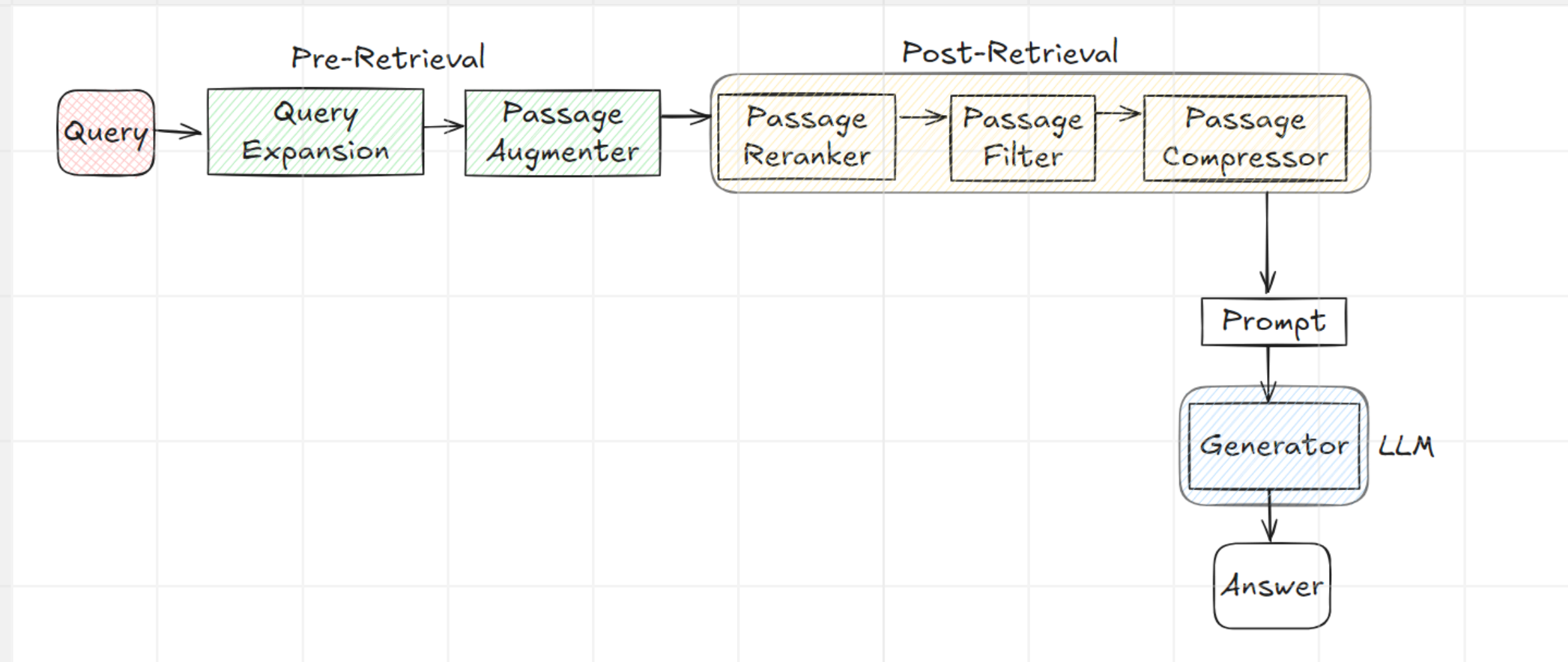
*This figure highlights how **ROUGE-SEM** captures both lexical similarity and semantic correctness, offering a more comprehensive evaluation compared to traditional ROUGE metrics.*[4]

Fig. 2. Typical examples of ROUGE-SEM on the DialSummEval dataset.

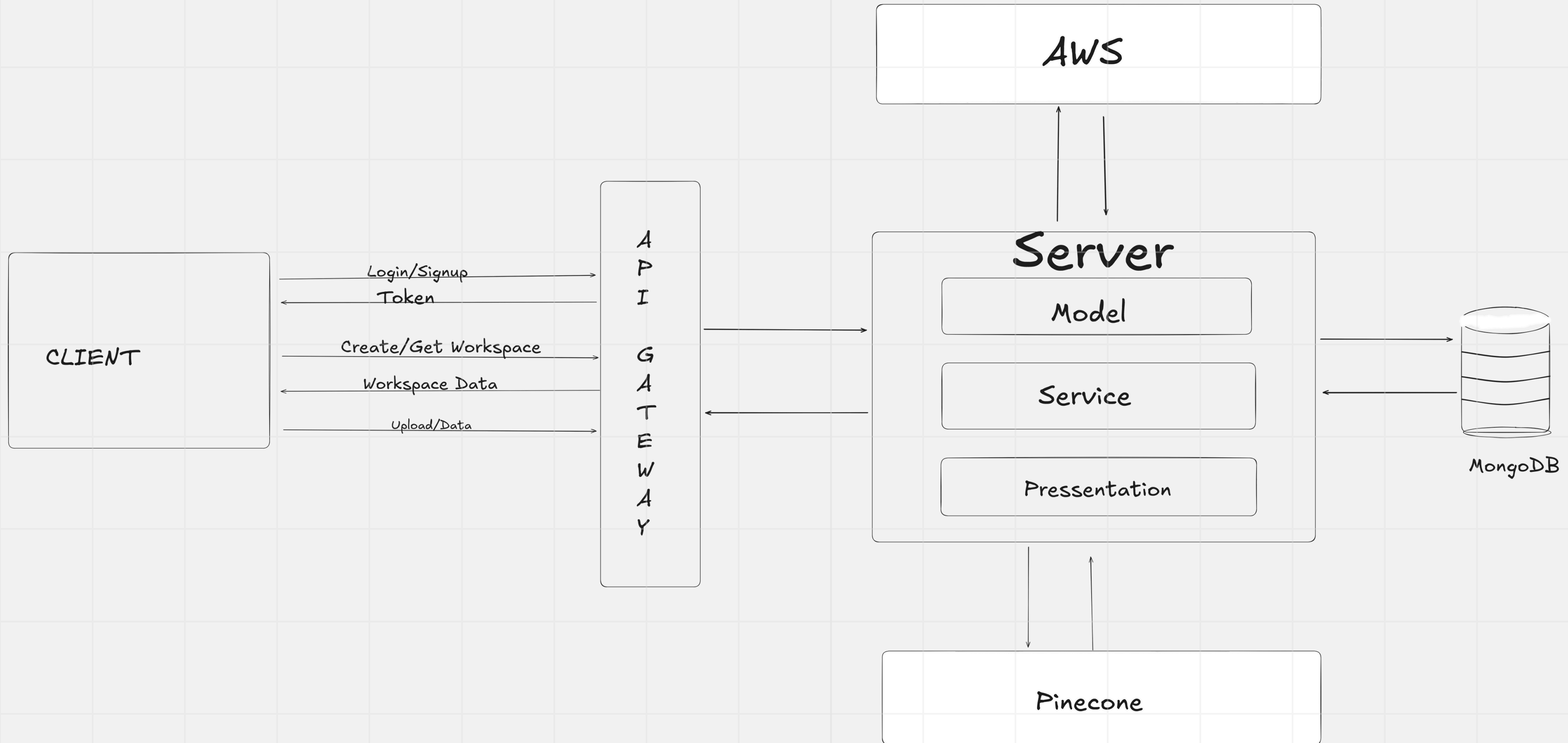
FLOWCHART



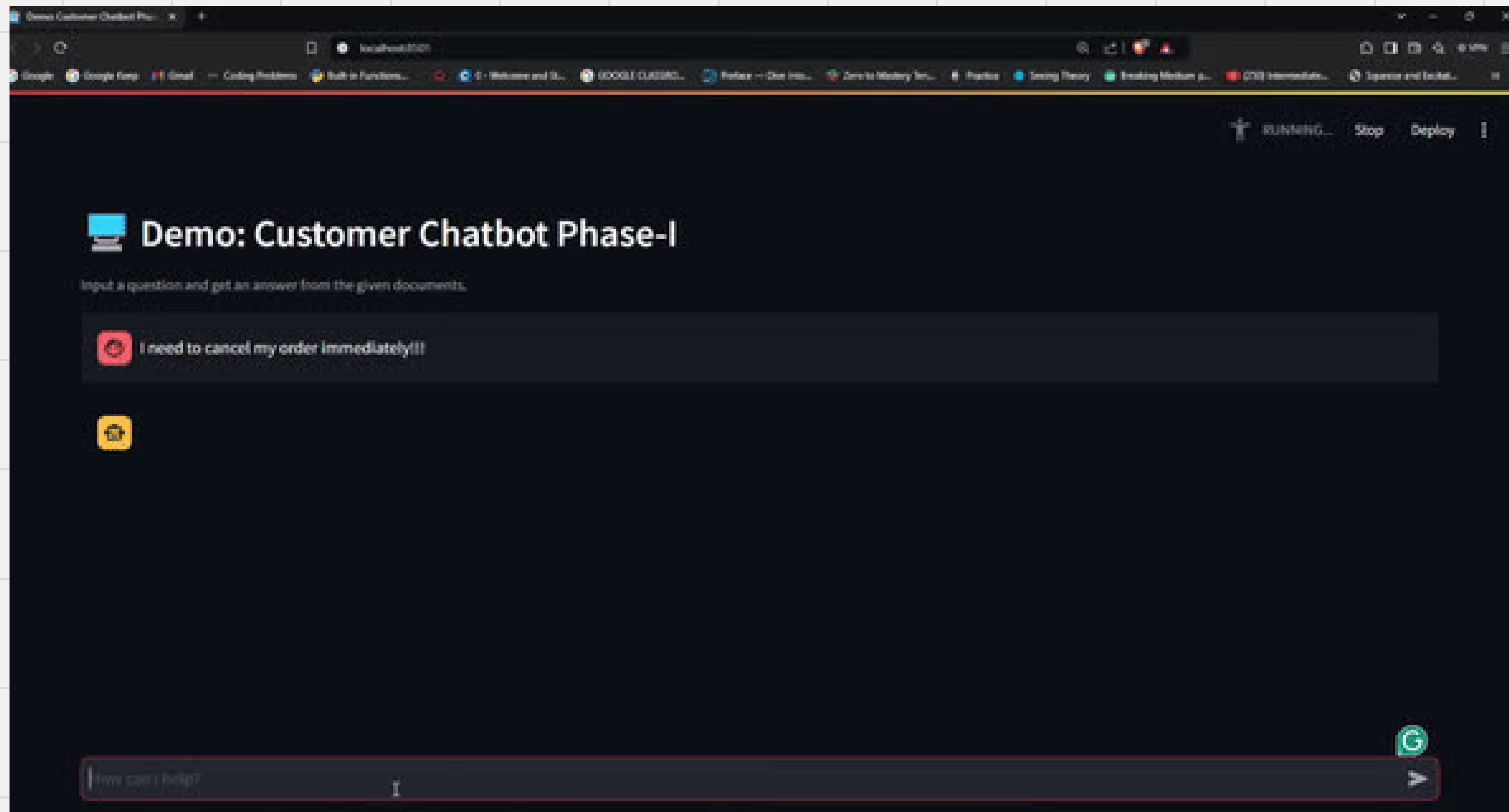
FINE TUNED RAG FUSION ARCHITECTURE



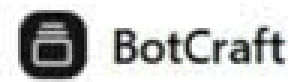
SYSTEM DESIGN



DEMO: PHASE-I



DEMO: PHASE-II



Login to your account

Enter your email below to login to your account

Email

m@example.com



Password



sayyedaribhussain321@gmail....



Manage addresses...

Login

Or continue with



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Don't have an account? [Sign up](#)

DEMO-II

Type your message...

bhussain@gmai...



TIMELINE (PHASE II)

STEP 1
**LITERATURE
SURVEY**

STEP 3
**DATA COLLECTION
AND PROCESSING**

STEP 5
**MODEL
EXPERIMENTATION**

STEP 7
**START BUILDING
GENERAL PURPOSE
CHATBOT BUILDER**

STEP 2
**PROBLEM
DEFINITION**

STEP 4
**RAG ARCHITECTURE
IMPLEMENTATION**

STEP 6
**INTEGRATION
AND TESTING**

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Q&A

THANK YOU